

# Geometric Score Function

## Breakdown of Appendix B.1: A General Framework for the Convergence of the Limiting Distribution

Instead of directly analysing the complex neural network scores right away, the authors first build a “general framework”. They analyse how any probability distribution of a specific mathematical form behaves as a scaling parameter (like noise) shrinks to zero.

### 1. The Setup: The Target Density Function

The authors consider a probability density that looks like this:  $\exp(-(f_\theta(x))/\theta)$ .

Here,  $\theta$  is a small parameter that will eventually go to zero (in the main text, this is related to the noise level  $\sigma$ ). The function  $f_\theta(x)$  is broken down into a Taylor-like expansion:

$$f_\theta(x) = f_0(x) + \theta f_1(x) + \hat{f}(x, \theta)$$

- $f_0(x)$ : The leading-order term. Its minimum sits exactly on the data manifold  $\mathcal{M}$ . This term represents the geometry.
- $f_1(x)$ : The next-order term. This term will eventually encode the actual data distribution on the manifold.
- $\hat{f}(x, \theta)$ : A tiny remainder that shrinks faster than  $\theta$ , so it won’t affect the final limit.

Because the density will eventually concentrate tightly around the manifold, they stop using standard ambient coordinates and switch to local coordinates  $(u, r)$ .

- $u$ : The coordinate moving tangentially to the manifold.
- $r$ : The coordinate moving orthogonally to the manifold.

### 2. Assumption B.1: The Ground Rules

Before doing calculus, they need to ensure the maths won’t break. Assumption B.1 sets these rules:

- The manifold  $\mathcal{M}$  is compact, smooth ( $C^4$ ), and has no boundaries.
- The leading term  $f_0$  is completely minimised exactly on the manifold.
- The functions are smooth enough to take multiple derivatives ( $f_0$  is  $C^3$ ,  $f_1$  is  $C^1$ ).
- **The Curvature Condition:** The second derivative (Hessian) of  $f_0$  in the perpendicular  $r$  direction is strictly positive. This means  $f_0$  curves sharply upward as you move away from the manifold, forming a steep valley that traps the probability mass.

### 3. Corollary B.1 & Lemma B.1: Laplace’s Method

The authors need to figure out what happens to the probability mass in the perpendicular  $r$  direction as the valley gets infinitely steep ( $\theta \rightarrow 0$ ). To do this, they rely on Laplace’s Method for Integrals.

- **Corollary B.1:** This is an adaptation of an existing mathematical theorem (Łapiński, 2019) that provides strict error bounds when approximating complex integrals of exponential functions.

- **Lemma B.1:** They apply Laplace’s method specifically to integrate out the normal coordinate  $r$ . By integrating over  $r$ , they squash the probability cloud flat onto the surface of the manifold.

The result of Lemma B.1 shows that integrating out  $r$  yields a formula dominated by  $\exp(-f_1(u, 0))$  and divided by the square root of the Hessian of  $f_0$ .

#### 4. Lemma B.2: Concentration of Measure

This lemma proves that as  $\theta$  approaches zero, the probability of finding a data point outside the manifold drops to exactly zero.

It formally proves that the density completely concentrates within the tubular neighbourhood and ultimately collapses onto the set of minimisers for  $f_0$ . If the resulting density converges to a stable distribution, this lemma guarantees that the support of that final distribution is strictly contained within the manifold  $\mathcal{M}$ .

#### 5. Theorem B.1: The Grand Conclusion

As  $\theta \rightarrow 0$ , the probability distribution  $\pi_\theta(x)$  converges weakly to a distribution  $\pi(u)$  on the manifold, defined as:

$$\pi(u) = \frac{\exp(-f_1(u, 0)) \left| \frac{\partial^2 f_0(u, 0)}{\partial r^2} \right|^{-1/2} d\mathcal{M}(u)/du}{Z}$$

(Note:  $Z$  represents the normalising integral in the denominator to ensure the probabilities sum to 1.)

**What this formula means:**

1.  **$\exp(-f_1(u, 0))$ :** The density on the manifold is entirely dictated by the higher-order term  $f_1$ . The leading term  $f_0$  just forced the points onto the manifold, but  $f_1$  decides how they are spread out across the surface.
2. **The Hessian Term:** This is a geometric correction factor (the inverse square root of the normal Hessian). It adjusts the probability density based on how sharply the space curves away from the manifold at that specific point  $u$ .
3. **Intrinsic Measure:** The  $d\mathcal{M}(u)/du$  ensures the maths respects the natural, non-Euclidean curvature of the manifold.

#### Line-by-Line Breakdown of Corollary B.1

At its core, Corollary B.1 is a rigorous statement of Laplace’s Method. Laplace’s method approximates integrals of the form  $\int g(x)e^{-F(x)/\theta}dx$  when  $\theta$  becomes very small. As  $\theta \rightarrow 0$ , the function  $e^{-F(x)/\theta}$  creates an incredibly sharp, peak precisely at the minimum of  $F(x)$ . Because the peak is so sharp, almost all the mass (the value of the integral) comes from the immediate neighbourhood around that minimum.

##### 1. Setting the Stage: The Space and the Functions

**Line:** “Let  $\Omega \subset \mathbb{R}^m$  be an open set and let  $\Omega' \subset \Omega$  be a closed ball. Let  $c_1 := \text{Vol}(\Omega')$ .”

- **The Maths:**  $\Omega$  is the entire mathematical space (or domain) we are integrating over.  $\Omega'$  is a smaller, tightly defined bubble (a closed ball) inside that space.

- **The Concept:** We know the action happens at the minimum of our function.  $\Omega'$  is defined as the specific neighbourhood immediately surrounding that minimum. We record its volume as  $c_1$  so we can use it to bound our error later.

**Line:** “Let  $F, g : \Omega \rightarrow \mathbb{R}$  with the following assumptions:”

- **The Concept:** We are dealing with two functions.  $F(x)$  sits in the exponent and drives the sharp peak.  $g(x)$  sits outside the exponent and modulates the height or amplitude of the function.

## 2. Assumptions on $F(x)$ (The Exponent)

**Line:**

$F|_{\Omega'} \in C^3(\Omega')$  and  $F \geq 0$  on  $\Omega$ . There is a unique minimiser  $x^* \in \text{int}(\Omega')$  of  $F$  on  $\Omega$ .

- **The Maths:** Inside our bubble  $\Omega'$ ,  $F$  is three-times continuously differentiable ( $C^3$ ).  $F$  is never negative, and it hits its bottom value at exactly one point,  $x^*$ , which is safely inside the bubble.
- **The Concept:** We need  $F$  to be highly smooth near the minimum so we can approximate it with a parabola (a Taylor expansion up to the second derivative).

**Line:**

$$m_1 := \inf_{x \in \Omega \setminus \Omega'} \{F(x) - F(x^*)\} > 0, \quad m_2 := \inf_{x \in \Omega'} \lambda_{\min}(\nabla^2 F(x)) > 0$$

(Note: The text briefly contains a typo calling the first term  $n_1$ , but later refers to it correctly as  $m_1$ ).

- **The Maths ( $m_1$ ):** This looks at all points *outside* our bubble. It states that the value of  $F(x)$  everywhere outside the bubble is strictly greater than the minimum  $F(x^*)$  by at least a gap of  $m_1$ . This guarantees there are no “competing” peaks elsewhere in the domain that could mess up the integral.
- **The Maths ( $m_2$ ):** Inside the bubble, we look at the Hessian matrix  $\nabla^2 F(x)$  (the matrix of second derivatives, which measures curvature). We require its smallest eigenvalue ( $\lambda_{\min}$ ) to be bounded away from zero by at least  $m_2$ . This guarantees the minimum is strictly “sharp” or convex—it looks like a steep bowl, not a flat trough.

**Line:** “Let  $c_2 := \sup_{x \in \Omega'} \|\nabla^2 F(x)\|$ ,  $c_3 := \sup_{x \in \Omega'} \|\nabla^3 F(x)\|$ ”

- **The Concept:** The authors are just taking inventory. They record the maximum possible values (the supremum) of the second ( $c_2$ ) and third ( $c_3$ ) derivatives inside the bubble. By capping how wildly  $F$  can curve, they can strictly bound the error of their approximation.

## 3. Assumptions on $g(x)$ (The Amplitude)

**Line:** “ $g|_{\Omega'} \in C^1(\Omega')$  and  $\int_{\Omega} |g(x)| dx < \infty$ ”

- **The Maths:** Inside the bubble,  $g$  must be smooth (differentiable once). Across the entire space, the absolute area under  $g(x)$  must be finite.
- **The Concept:** If  $g(x)$  were allowed to explode to infinity somewhere outside the bubble, it could overpower the exponential decay of  $F(x)$  and ruin the integral.

**Line:** “Let  $c_4 := \sup_{x \in \Omega'} |g(x)|$ ,  $c_5 := \sup_{x \in \Omega'} \|\nabla g(x)\|$ ,  $c_6 := \int_{\Omega} |g(x)| dx$ .”

- **The Concept:** Again, taking inventory. They record the maximum height of  $g$  ( $c_4$ ), the maximum slope of  $g$  ( $c_5$ ) inside the bubble, and the total area of  $g$  across the whole space ( $c_6$ ).

#### 4. The Grand Result: The Integral Approximation

**Line:** “Then, for every  $\theta > 0$ .”

$$\int_{\Omega} g(x) e^{-F(x)/\theta} dx = \exp(-F(x^*)/\theta) \frac{(2\pi\theta)^{m/2}}{\sqrt{|\nabla^2 F(x^*)|}} (g(x^*) + h(\theta))$$

This is the core equation of the corollary. Because the peak is so sharp around  $x^*$ , we can evaluate the integral by effectively replacing  $F(x)$  with a parabolic Taylor expansion around  $x^*$ , turning the integrand into a standard Gaussian curve. Here are the pieces:

1.  $\exp(-F(x^*)/\theta)$ : The raw height of the exponential function exactly at its peak.
2. **The Volume Term:** The term with  $(2\pi\theta)^{m/2}$  over the square root of the Hessian determinant represents the volume or “width” of the peak. When you integrate a Gaussian, the area depends on the variance. Here,  $\theta$  acts like the variance. The sharper the bowl, the smaller the volume under the peak.
3.  $g(x^*)$ : Because the peak is infinitesimally narrow, the integral essentially only “sees” the function  $g(x)$  at the exact point  $x^*$ .
4.  $h(\theta)$ : The mathematical error of this approximation.

#### 5. Bounding the Error

**Line:** “where  $|h(\theta)|$  can be upper bounded by a function of  $(c_1, \dots, c_6, m_1, m_2)$ . Moreover,  $h(\theta) = O(\sqrt{\theta})$  as  $\theta \rightarrow 0$ .”

- **The Maths:** The error  $h(\theta)$  shrinks proportionally to  $\sqrt{\theta}$  as  $\theta$  gets closer to zero.
- **The Concept:** Because they carefully logged all those boundaries ( $c_1$  through  $c_6$ , and  $m_1, m_2$ ), they can guarantee that the error won’t randomly blow up. It is strictly controlled by those specific parameters.

**Line:** “The  $O(\sqrt{\theta})$  is uniform over any class of pairs  $(F, g)$  for which  $c_1, \dots, c_6$  are bounded above and  $m_1, m_2$  are bounded below by strictly positive constants uniformly over the class.”

- **The Concept:** This ensures that this approximation works consistently (uniformly) for *any* functions  $F$  and  $g$  you might swap in, as long as they obey the minimum and maximum boundaries established earlier.

#### Line-by-Line Breakdown of Lemma B.1

**Goal:** The primary purpose of Lemma B.1 is to take a high-dimensional probability distribution that is concentrated near a data manifold and “flatten” it onto the surface of that manifold. It does this mathematically by integrating out the normal (perpendicular) coordinates.

##### 1. The Setup and the Goal

**Line:** “Assume Assumption B.1, and let  $h(x) : \mathbb{R}^d \rightarrow \mathbb{R}$  be  $C^1$  and uniformly bounded in  $T_{\mathcal{M}}(\epsilon)$ . Define  $h(u, r) := h(\Phi(u, r))$ .”

- **The Maths:** We are bringing in the safety rules from Assumption B.1 (smoothness, compact manifold, sharp minimums). We also introduce a new function,  $h(x)$ , which is smooth ( $C^1$ ) and doesn’t blow up to infinity inside our tubular neighbourhood  $T_{\mathcal{M}}(\epsilon)$ . We then map it into our local coordinates  $(u, r)$ .

- **The Concept:** Think of  $h(x)$  as a generic “test function” or an arbitrary amplitude. The authors are setting up a general integral so this lemma can be reused flexibly later in the paper.

**Line:** “Then we have”

$$\int_{\|r\|<\epsilon} \exp\left(-\frac{f_\theta(u, r)}{\theta}\right) h(u, r) dr = \exp\left(-\frac{f_0(u, 0)}{\theta}\right) \exp(-f_1(u, 0)) \frac{(2\pi\theta)^{(d-n)/2}}{\sqrt{\left|\frac{\partial^2 f_0}{\partial r^2}(u, 0)\right|}} (h(u, 0) + o(1))$$

“where the  $o(1)$  term is uniform for  $u$ .”

- **The Concept:** This is the destination. They are integrating *only* over  $r$  (the normal/perpendicular direction) within the tiny distance  $\epsilon$  from the manifold. Because the probability forms a sharp peak right on the manifold (where  $r = 0$ ), integrating across the peak effectively collapses the equation down to evaluate everything exactly at  $r = 0$ .

## 2. The Proof: Step 1 - Unpacking the Exponent

**Line:** “We have that”

$$\int_{\|r\|<\epsilon} \exp\left(-\frac{f_\theta(u, r)}{\theta}\right) h(u, r) dr = \int_{\|r\|<\epsilon} \exp\left(-\frac{f_0(u, r)}{\theta}\right) \exp(-f_1(u, r)) h(u, r) \exp\left(-\frac{\hat{f}(u, r, \theta)}{\theta}\right) dr$$

- **The Maths:** They take the target function  $f_\theta$  and expand it into its three known parts:  $f_0$ ,  $f_1$ , and  $\hat{f}$ .
- **The Concept:** They are isolating the dominant geometric term ( $f_0$ ), the density term ( $f_1$ ), and the annoying perturbation/error term ( $\hat{f}$ ).

## 3. The Proof: Step 2 - The Plus-One / Minus-One Trick

**Line:**

$$\begin{aligned} &= \int_{\|r\|<\epsilon} \exp\left(-\frac{f_0(u, r)}{\theta}\right) \exp(-f_1(u, r)) h(u, r) dr \\ &\quad + \int_{\|r\|<\epsilon} \exp\left(-\frac{f_0(u, r)}{\theta}\right) \exp(-f_1(u, r)) h(u, r) \left(\exp\left(-\frac{\hat{f}(u, r, \theta)}{\theta}\right) - 1\right) dr \end{aligned}$$

- **The Maths:** Look at the last term from the previous step:  $\exp(-\hat{f}/\theta)$ . They rewrite this as  $(1 + [\exp(-\hat{f}/\theta) - 1])$ . They then multiply this through the integral to split it into two separate integrals.
- **The Concept:** The first integral is now completely “clean”—it only contains  $f_0$ ,  $f_1$ , and  $h$ . The second integral acts as a “bin” that captures all the messy error associated with  $\hat{f}$ .

## 4. The Proof: Step 3 - Solving the Clean Integral

**Line:** “For the first term, we can directly apply Corollary B.1 with  $F(r) = f_0(u, r)$ ,  $g(r) = \exp(-f_1(u, r))h(u, r)$ , and  $\Omega'$  being the ball  $\{r \mid \|r\| \leq \hat{\epsilon}\}$ .”

- **The Concept:** They match the pieces of their clean integral to the variables required by Laplace’s method.

**Line:** “Define”

$$J = \exp\left(-\frac{f_0(u, 0)}{\theta}\right) \exp(-f_1(u, 0)) \frac{(2\pi\theta)^{(d-n)/2}}{\sqrt{\left|\frac{\partial^2 f_0}{\partial r^2}(u, 0)\right|}}$$

“The first term can be approximated as  $J(h(u, 0) + o(1))$ ”

- **The Maths:** By plugging the pieces directly into the formula from Corollary B.1, they get the exact result they are looking for.

## 5. The Proof: Step 4 - Trashing the Bin Integral

**Line:** “The second term can be upper bounded by”

$$\sup_r |h(u, r)| \cdot \sup_r \left| \exp\left(-\frac{\hat{f}(u, r, \theta)}{\theta}\right) - 1 \right| \int_{\|r\| < \epsilon} \exp\left(-\frac{f_0(u, r)}{\theta}\right) \exp(-f_1(u, r)) dr$$

- **The Maths:** They find the absolute maximum possible size of the second (bin) integral by pulling the amplitude and the error factor outside the integral via the supremum (sup).

**Line:**

$$= o(1)J(1 + o(1)) = o(1)J$$

“where we used Corollary B.1 for the integral. The lower bound can be obtained similarly. The result follows.”

- **The Maths:** By Assumption B.1, we know  $\hat{f}$  shrinks faster than  $\theta$ . Therefore,  $\frac{\hat{f}}{\theta} \rightarrow 0$ . As a result,  $\exp(0) - 1 = 1 - 1 = 0$ . So that entire supremum term becomes vanishingly small ( $o(1)$ ).

## Line-by-Line Breakdown of Lemma B.2

**Goal:** This lemma formally proves “Concentration of Measure.” It demonstrates that as the noise parameter ( $\theta$ ) shrinks to zero, the probability distribution completely collapses onto the data manifold  $\mathcal{M}$ .

### 1. The Setup and Assumptions

**Line:** “Let  $f_\theta(x) = f_0(x) + \bar{f}(x, \theta)$ , such that  $\exp(-f_\theta(x)/\theta)$  is a normalised density function on  $\mathbb{R}^d$ .”

- **The Maths:** They define the exponent of the probability density as having two parts: a dominant geometric term  $f_0(x)$  and an error/perturbation term  $\bar{f}(x, \theta)$ .

**Line:** “Suppose  $\mathcal{M}$  is a connected and compact  $C^4$  manifold without boundary. Assume that:

1.  $f_0(x)$  is continuous with  $\arg \min_{x \in T_{\mathcal{M}}(\epsilon)} f_0(x) = \mathcal{M}$  and  $\min_{x \in T_{\mathcal{M}}(\epsilon)} f_0(x) = 0$ .”

- **The Maths:** The dominant function  $f_0$  reaches its absolute minimum (which is exactly zero) *only* when evaluated exactly on the manifold  $\mathcal{M}$ .

**Line:** “2.  $\bar{f}(x, \theta)$  is continuous and uniformly  $o(1)$  as  $\theta \rightarrow 0$  for all  $x \in \overline{T_{\mathcal{M}}(\epsilon)}$ .”

- **The Maths:** The error term  $\bar{f}$  vanishes completely (goes to  $o(1)$ ) as  $\theta$  goes to zero.

**Line:** “3. The density concentrates in  $T_{\mathcal{M}}(\epsilon)$ , i.e.,”

$$\lim_{\theta \rightarrow 0} \int_{T_{\mathcal{M}}(\epsilon)} \exp\left(-\frac{f_{\theta}(x)}{\theta}\right) dx = 1$$

- **The Maths:** 100% of the probability mass will eventually end up somewhere inside the tubular neighbourhood  $(T_{\mathcal{M}}(\epsilon))$ .

## 2. The Core Claim

**Line:** “For any  $\eta > 0$ , define the set  $C_{\eta} = \{x \mid f_0(x) > \eta\}$ . Then,”

$$\int_{C_{\eta} \cup T_{\mathcal{M}}(\epsilon)^c} \exp\left(-\frac{f_{\theta}(x)}{\theta}\right) dx \rightarrow 0 \text{ as } \theta \rightarrow 0$$

- **The Maths:** They define a region  $C_{\eta}$  which is everywhere that  $f_0(x)$  is greater than some tiny positive number  $\eta$ . They claim the probability of finding a data point in this region shrinks to exactly zero.

**Line:** “If in addition,  $\exp(-f_{\theta}(x)/\theta)$  converges weakly to a distribution as  $\theta \rightarrow 0$ , the support of the limiting distribution is contained in  $\mathcal{M}$ .”

## 3. The Proof: Bounding the Probability

**Line:** “Proof. Since we have that  $\int_{T_{\mathcal{M}}(\epsilon)} \exp(-f_{\theta}(x)/\theta) dx \rightarrow 1$  for the first result, it suffices to show that  $\int_{T_{\mathcal{M}}(\epsilon) \cap C_{\eta}} \exp(-f_{\theta}(x)/\theta) dx \rightarrow 0$ .”

**Line:** “According to the assumptions, we have that for any  $\delta > 0$ ,  $\exists \theta_0$ , such that  $\forall \theta < \theta_0$ ,  $|\bar{f}(x, \theta)| < \delta$ . Therefore, we have”

$$\int_{T_{\mathcal{M}}(\epsilon) \cap C_{\eta}} \exp\left(-\frac{f_{\theta}(x)}{\theta}\right) dx \leq \int_{T_{\mathcal{M}}(\epsilon) \cap C_{\eta}} \exp\left(\frac{-\eta + \delta}{\theta}\right) dx \leq \text{Vol}(T_{\mathcal{M}}(\epsilon)) \exp\left(\frac{-\eta + \delta}{\theta}\right)$$

- **The Maths:** Since the error term  $\bar{f}$  vanishes, they bound its maximum absolute value by  $\delta$ . In the region where  $f_0 > \eta$ , they bound the exponent by its worst-case scenario:  $(-\eta + \delta)/\theta$ .

**Line:** “We choose  $\delta = \eta/2$  then the right-hand side goes to zero as  $\theta \rightarrow 0$ .”

- **The Concept:** If  $\delta$  is chosen to be half the size of  $\eta$ , then  $(-\eta + \delta)$  equals  $-\eta/2$ , which is a strictly negative number. The limit of  $\exp(-\eta/(2\theta))$  as  $\theta \rightarrow 0$  approaches 0.

## 4. The Proof: Weak Convergence

**Line:** “Let the limiting measure be  $P$ , and  $P_{\theta}$  be the probability measure corresponding to the density  $\exp(-f_{\theta}(x)/\theta)$ . Since  $C_{\eta}$  is an open set, we have that”

$$P(C_{\eta}) \leq \liminf_{\theta \rightarrow 0} P_{\theta}(C_{\eta}) = 0$$

**Line:** “Denote  $C := \mathcal{M}^c$ . We have that  $C = \cup_{m=1}^{\infty} C_{1/m} \cup \overline{T_{\mathcal{M}}(\epsilon)}^c$ . Then we have”

$$P(C) \leq \sum_{m=1}^{\infty} P(C_{1/m}) + P(\overline{T_{\mathcal{M}}(\epsilon)}^c) = 0$$

“which concludes the proof.”

## Line-by-Line Breakdown of Theorem B.1

**Goal:** This theorem provides the exact, explicit formula for the limiting probability distribution on the data manifold.

### 1. The Setup and Assumptions

**Line:** “Theorem B.1. Assume Assumption B.1. Define  $\pi_\theta(x) \propto \exp\left(-\frac{f_\theta(x)}{\theta}\right)$ ,”

**Line:** “Assume that  $1 - \int_{x \in T_{\mathcal{M}}(\epsilon)} \pi_\theta(x) dx \rightarrow 0$  as  $\theta \rightarrow 0$ .”

### 2. The Core Claim and Formula

**Line:** “Then we have that as  $\theta \rightarrow 0$ ,  $\pi_\theta$  converges weakly to the following distribution:”

$$\pi(u) = \frac{\exp(-f_1(u, 0)) \left| \frac{\partial^2 f_0(u, 0)}{\partial r^2} \right|^{-1/2} d\mathcal{M}(u)/du}{\int_{\mathcal{M}} \exp(-f_1(u, 0)) \left| \frac{\partial^2 f_0(u, 0)}{\partial r^2} \right|^{-1/2} d\mathcal{M}(u)/du}$$

- **The Maths:** This is the formula for the limiting distribution.

- **The Breakdown of the Numerator:**

- $\exp(-f_1(u, 0))$ : The density is driven by the higher-order term  $f_1$ .
- **Geometric Correction:** The term involving the inverse square root of the normal Hessian adjusts the probability density based on how sharply the ambient space curves into the manifold.

**Line:** “where  $d\mathcal{M}$  is the intrinsic measure on the manifold  $\mathcal{M}$ , i.e.,  $d\mathcal{M}(u) = |g(u)|^{1/2} du$ , and  $du$  is the Lebesgue measure on the local parameterisation domain  $U$ .”

### 3. The Proof Strategy

**Line:** “Proof. The proof follows the same as the proof in Hwang (1980, Theorem 3.1). The only difference is that we replace the estimate of Hwang (1980, Equation (3.2)) with our Lemma B.1.”

**Line:** “Note that the  $Q$  in Hwang (1980, Theorem 3.1) is assumed as a probability measure, thus  $f$  (in his notation) integrates to one. However, the proof technique of Hwang (1980, Theorem 3.1) remains valid even if  $f$  is not a probability density, so applying to our case.  $\square$ ”

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## Line-by-Line Breakdown of Theorem 3.1

**Goal:** This theorem mathematically proves the “separation of scales”—demonstrating precisely why it is mathematically easier for a model to learn the geometric shape of the manifold than to learn the actual data distribution upon it.

### 1. The Setup and Assumptions

**Line:** “Theorem 3.1 (Informal Theorem B.2). Assume Assumptions 2.1 and 2.2 holds. For any  $x \in T_{\mathcal{M}}(\epsilon)$ ,”

## 2. The Core Equation

Line:

$$\log p_\sigma(x) = -\frac{1}{\sigma^2}d_{\mathcal{M}}(x) + \log p_{data}(\Phi^{-1}(P_{\mathcal{M}}(x))) - \frac{d-n}{2}\log(2\pi\sigma^2) + H(x) + o(1)$$

This equation deconstructs the log-probability density of the noisy data ( $\log p_\sigma(x)$ ) into five distinct components:

- **Term 1: The Geometric Pull** ( $-\frac{1}{\sigma^2}d_{\mathcal{M}}(x)$ ): Because it is divided by  $\sigma^2$ , as  $\sigma \rightarrow 0$ , this term explodes toward  $-\infty$  for any point not perfectly resting on the manifold. This acts as a massive mathematical gravity well. Because it scales by  $\sigma^{-2}$ , it is the “loudest” and most dominant signal for the model to learn.
- **Term 2: The Data Distribution** ( $\log p_{data}(\Phi^{-1}(P_{\mathcal{M}}(x)))$ ): Crucially, there is no  $\sigma$  in this term. It is  $\Theta(1)$ , meaning its strength remains constant regardless of the noise level.
- **Term 3: The Perpendicular Volume** ( $-\frac{d-n}{2}\log(2\pi\sigma^2)$ ): This acts as the normalisation factor for the Gaussian noise.
- **Term 4: The Curvature** ( $H(x)$ ): This term captures the intrinsic geometric curvature of the manifold at that location.
- **Term 5: The Vanishing Error** ( $o(1)$ ): A mathematically tiny remainder that uniformly shrinks to zero.

## 3. The Grand Implication

This establishes a fundamental rate separation. Before a neural network can learn anything about the specific data distribution  $p_{data}$  (which requires extreme precision), it is forced to first recover the manifold geometry  $d_{\mathcal{M}}$ . Any error in learning the geometry is magnified by that  $\sigma^{-2}$  factor, completely drowning out the  $\Theta(1)$  distribution details.